

```

+=====+
I 0 0 0 0 0 0 0 0 | 0 0 I
I | | | | | | | | | | I
I | | | | | | | 0 | | I
+=====+
I | | | | | | | | | 0 I
I | | | | | | | | | 0 I
I 0 0 0 0 0 0 0 0 0 0 0 I
I 0 0 0 0 0 0 0 0 0 0 | I
I 0 0 0 0 0 0 0 0 0 0 | I
I 0 0 0 0 0 0 0 0 0 0 0 I
+==0==0==0==0==0==0==5==0==3==+
  +q w e r t y u i o p
  -a s d f g h j k l ;
(Enter a number, "quit", or a stream of up/down letters.)
--snip--

```

How It Works

The `displayAbacus()` function accepts a number argument used to figure out where it should render beads on the abacus. The soroban always has exactly 80 possible locations for either '0' beads or '|' rod segments, as marked by the curly braces ({} in the multiline string on lines 127 to 139. Another 10 curly braces represent the digits of the number argument.

We need to create a list of strings to fill in these curly braces, going from left to right, top to bottom. The code in `displayAbacus()` will populate a `hasBead` list with a `True` value to display a '0' bead and a `False` value to display a '|'. The first 10 values in this list are for the top “heaven” row. We’ll put a bead in this row if the column’s digit is 0, 1, 2, 3, or 4, since the heaven bead won’t be in that row unless the digit for that column is 0 to 4. We add Boolean values to `hasBead` for the remaining rows.

Lines 118 to 123 use `hasBead` to create an `abacusChar` list that contains the actual '0' and '|' strings. When combined with `numberList` on line 126, the program forms a `chars` list that populates the curly braces ({} for the multiline-string ASCII art of the soroban.

```

1. """Soroban Japanese Abacus, by Al Sweigart al@inventwithpython.com
2. A simulation of a Japanese abacus calculator tool.
3. More info at: https://en.wikipedia.org/wiki/Soroban
4. View this code at https://nostarch.com/big-book-small-python-projects
5. Tags: large, artistic, math, simulation"""
6.
7. NUMBER_OF_DIGITS = 10
8.
9.
10. def main():
11.     print('Soroban - The Japanese Abacus')
12.     print('By Al Sweigart al@inventwithpython.com')
13.     print()
14.
15.     abacusNumber = 0 # This is the number represented on the abacus.

```